

A SECOND ARITHMETIC PROGRESSION OF MATHIEU M_{23} -POLYNOMIALS

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ABSTRACT. Huang, Jackson, Lee, Poonen, Pries, and Zhang constructed an explicit regular M_{23} -extension of $\mathbf{Q}(T)$ using a degree-23 polynomial of parameter degree 8. We prove that every integer specialization in the arithmetic progression

$$T \equiv 83 \pmod{85}$$

has Galois group exactly M_{23} after taking the primitive part. The proof uses two reductions that are constant throughout the progression: modulo 17 the polynomial is irreducible of degree 23, while modulo 5 its irreducible-factor degrees are $(1, 2, 4, 8, 8)$. The resulting Galois group contains elements of orders 23 and 8. Since it embeds into the generic M_{23} and the only proper maximal subgroup of M_{23} whose order is divisible by 23 is the odd-order group $23:11$, the specialization group must be M_{23} . An exact PARI/GP certificate verifies the coefficient table, both frozen reductions, squarefreeness, and the group-order exclusion. This is a second explicit progression for the same regular cover, not a progression arising from an independent M_{23} -cover.

1. INTRODUCTION

Huang, Jackson, Lee, Poonen, Pries, and Zhang constructed a regular M_{23} -extension of $\mathbf{Q}(T)$ and supplied an exact defining polynomial $F(V, T)$ of degree 23 in V [1]. Put

$$\Phi(T, V) = (T^2 + 23)^4 F(V, T) \in \mathbf{Z}[T, V].$$

Its splitting field over $\mathbf{Q}(T)$ has Galois group M_{23} in its natural degree-23 action.

A companion paper [2] computes a minimal parameter-degree model of this same cover and proves that its primitive specializations have Galois group M_{23} whenever $T \equiv 2 \pmod{319}$. The purpose of the present note is to record a different uniform certificate, directly in the published degree- $(23, 8)$ equation.

Theorem 1.1 (Second progression). *For every $n \in \mathbf{Z}$, let $f_n(V)$ be the primitive part in $\mathbf{Z}[V]$ of*

$$\Phi(83 + 85n, V).$$

Then f_n has degree 23, is separable and irreducible over $\mathbf{Q}$, and

$$\text{Gal}(f_n/\mathbf{Q}) \cong M_{23}$$

in its natural action on the roots.

The argument is deliberately elementary once the generic M_{23} theorem is known. The modulus $85 = 5 \cdot 17$ freezes two squarefree Frobenius factorizations. One supplies a 23-cycle and the other an element of order 8; the ATLAS maximal-subgroup list then leaves no proper possibility.

The contribution is the explicit congruence class and its uniform exact certificate. General results on arithmetic progressions in Hilbert sets are considerably broader; no new general specialization theorem is asserted. Likewise, the present progression and the progression of [2] come from two source coordinates on the same regular cover. They do not constitute two independent M_{23} families.

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2. THE SPECIALIZATION SUBGROUP

We recall the standard passage from generic to specialized Galois groups in the form used below.

Lemma 2.1 (Specialization subgroup). *Let $Q(T, x) \in \mathbf{Q}[T, x]$ have splitting field $L/\mathbf{Q}(T)$ with Galois group G . Suppose that at $T = t \in \mathbf{Q}$ the leading coefficient and discriminant of Q with respect to x are nonzero. Then the splitting field of $Q(t, x)$ over $\mathbf{Q}$ has Galois group isomorphic, in its action on the specialized roots, to a subgroup of G .*

Proof. Work over the discrete valuation ring $R = \mathbf{Q}[T]_{(T-t)}$. After dividing by the leading coefficient, the polynomial is monic with unit discriminant. The normalization of R in L is therefore unramified above $T = t$. Choose a prime above $(T - t)$. Its decomposition group maps isomorphically to the Galois group of the residue extension because the inertia group is trivial. The generic roots have distinct residues, and those residues are exactly the roots of $Q(t, x)$, so the residue extension is its splitting field. The decomposition group is a subgroup of G . $\square$

For our equation, the generic group G in Lemma 2.1 is M_{23} by [1]. It remains to force enough elements into every specialized subgroup.

3. THE TWO FROZEN REDUCTIONS

Let $c_n > 0$ be the greatest common divisor of the coefficients of $\Phi(83 + 85n, V)$, and set

$$f_n(V) = c_n^{-1} \Phi(83 + 85n, V).$$

Because 85 is divisible by both 5 and 17, coefficientwise reduction gives

$$\Phi(83 + 85n, V) \equiv \Phi(83, V) \pmod{5}, \quad \Phi(83 + 85n, V) \equiv \Phi(83, V) \pmod{17}.$$

The right-hand sides have degree 23 and are nonzero, so neither 5 nor 17 divides c_n . Passing to f_n consequently changes either reduction only by a nonzero scalar.

For a prime $p \in \{5, 17\}$, let $\bar{f}_{n,p}$ denote the monic scalar multiple of the reduction of f_n . Exact factorization over $\mathbf{F}_5$ gives

$$\begin{aligned} \bar{f}_{n,5} = & V(V^2 + 3V + 3)(V^4 + 2V^3 + V^2 + 2V + 3) \\ & \cdot (V^8 + V^7 + 2V^6 + V^4 + 4V^2 + V + 3) \\ & \cdot (V^8 + 4V^7 + 4V^6 + 2V^5 + 4V^4 + V^3 + 3V^2 + 2). \end{aligned}$$

All five displayed factors are irreducible and distinct. Thus the factor degrees are

$$(1, 2, 4, 8, 8),$$

and their least common multiple is 8.

Modulo 17, the monic reduction is

$$\begin{aligned} \bar{f}_{n,17} = & V^{23} + 16V^{21} + 15V^{20} + 8V^{19} + 3V^{18} + 15V^{17} + 5V^{16} + 2V^{15} \\ & + 14V^{14} + 15V^{13} + 5V^{12} + 5V^{11} + 6V^{10} + 12V^8 + 7V^7 + 10V^6 \\ & + 13V^5 + 11V^4 + 8V^3 + 8V^2 + 15V + 3. \end{aligned}$$

It is irreducible over $\mathbf{F}_{17}$. The accompanying certificate checks this both by exact factorization and by the prime-degree Rabin criterion. In particular, both displayed reductions are squarefree.

Proposition 3.1 (Uniform Frobenius witnesses). *For every $n \in \mathbf{Z}$, the Galois group of f_n contains a 23-cycle and an element of cycle shape $(1, 2, 4, 8, 8)$, hence an element of order 8.*

Proof. The reductions above are independent of n , retain degree 23, and are squarefree. The Dedekind–Frobenius theorem identifies their irreducible factor degrees with cycle lengths of elements in the characteristic-zero Galois group. The irreducible reduction modulo 17 gives a 23-cycle; the reduction modulo 5 gives the stated cycle shape and order. $\square$

4. PROOF OF THE PROGRESSION THEOREM

Proof of Theorem 1.1. The degree-23 reduction modulo 17 shows first that every f_n has degree 23. Squarefreeness modulo either witness prime implies that its characteristic-zero discriminant is nonzero, so Lemma 2.1 embeds

$$H_n := \text{Gal}(f_n/\mathbf{Q}) \hookrightarrow M_{23}.$$

By Proposition 3.1, H_n contains elements of orders 23 and 8.

If H_n were proper, it would lie in a maximal subgroup of M_{23} . The ATLAS list [3] gives the maximal-subgroup orders

$$443520, \quad 40320, \quad 40320, \quad 20160, \quad 7920, \quad 5760, \quad 253.$$

Only $253 = 23 \cdot 11$ is divisible by 23. Therefore a proper H_n containing an element of order 23 would lie in the subgroup 23:11. That subgroup has odd order and cannot contain the element of order 8 already present in H_n . Hence $H_n = M_{23}$.

The 23-cycle also acts transitively on the roots, so f_n is irreducible over $\mathbf{Q}$. This completes the proof. $\square$

The two known progressions for this cover intersect explicitly.

Corollary 4.1 (Common subprogression). *The simultaneous congruences*

$$T \equiv 2 \pmod{319}, \quad T \equiv 83 \pmod{85}$$

are equivalent to

$$T \equiv 22013 \pmod{27115}.$$

Thus every integer in this common subprogression satisfies both uniform two-prime certificates, applied to their respective equations.

Proof. The moduli $319 = 11 \cdot 29$ and $85 = 5 \cdot 17$ are coprime. The Chinese remainder theorem gives one class modulo $319 \cdot 85 = 27115$; direct substitution gives the representative 22013. $\square$

Remark 4.2. The conclusion of Corollary 4.1 is a comparison of two certificates for the same generic cover. It does not supply a second geometrically inequivalent M_{23} -cover.

5. EXACT CERTIFICATION AND SCOPE

The complete integral coefficient table for Φ is stored as

`data/Fint_coefficients_Z.gp.`

It was generated from the published model of [1]; its SHA-256 digest is

`de667c8a260544493d993d83443095b8a27dda1d170ab521bcd3373d1a403d39.`

The PARI/GP program

`verification/verify_progression_85.gp`

checks the degree and content at $T = 83$, the two displayed factorizations, the symbolic congruences after substituting $T = 83 + 85n$, squarefreeness, Rabin irreducibility modulo 17, and the maximal-subgroup order exclusion. From the repository root, the entire certificate and paper are reproduced by

`make verify, \quad make paper.`

The regular M_{23} theorem and the degree- $(23, 8)$ equation are due to Huang–Jackson–Lee–Poonen–Pries–Zhang. The present contribution is the explicit progression $83 \pmod{85}$ and its uniform certificate. The argument does not prove that specializations with different values of n are linearly disjoint, does not classify their ramification, and makes no claim about zeta functions or Gaussian multiplicative chaos.

The discovery and drafting process used substantial AI assistance.

REFERENCES

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